



ML FOR SCIENCE · LECTURE 1

Introduction & the SciML Landscape

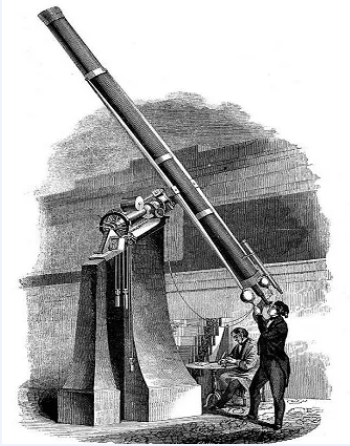
What is **machine learning for science**, and how does it change the way we **model, simulate, and discover the natural world**?

Four Paradigms of Science

Empirical

(thousand years ago)

Observation & experiment
(Kepler, Galileo)



Theoretical

(last few hundred years)

Laws & closed-form models
(Newton, Maxwell)

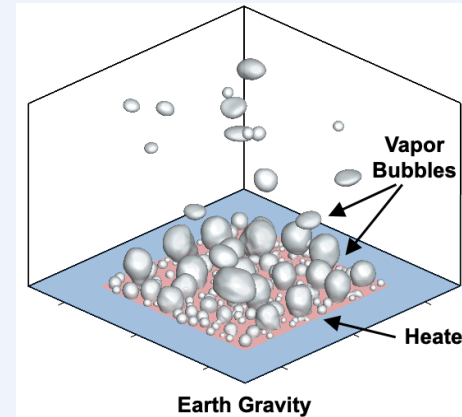
$$F = G \frac{m_1 m_2}{r^2}$$



Computational

(last few decades)

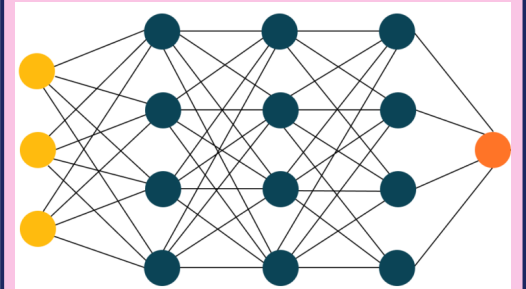
Simulating the theory
(numerical PDE solvers)



Data-driven

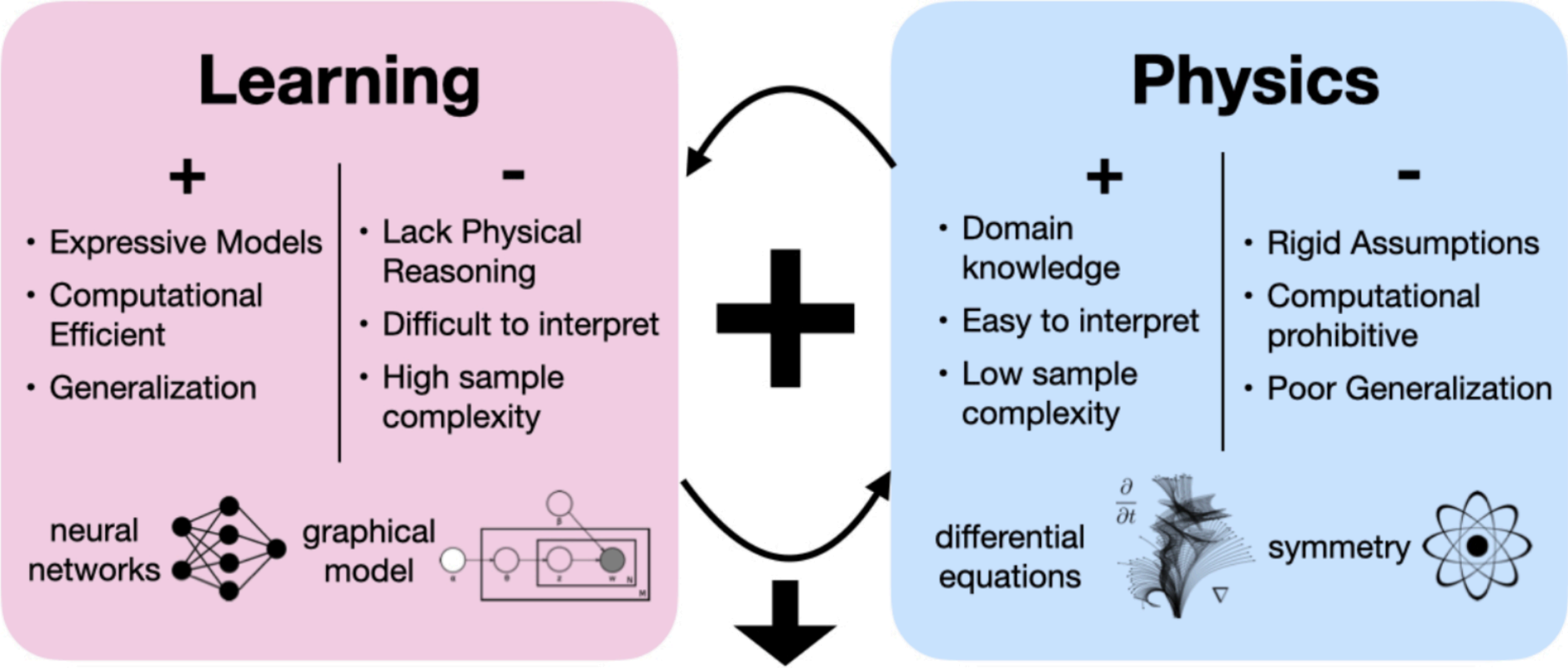
(today)

Learning from data at scale
(scientific ML)



ML for science doesn't replace the first three paradigms, it accelerates and augments them, often unifying theory, experiments, and simulation

Two Branches of Computing, Converging



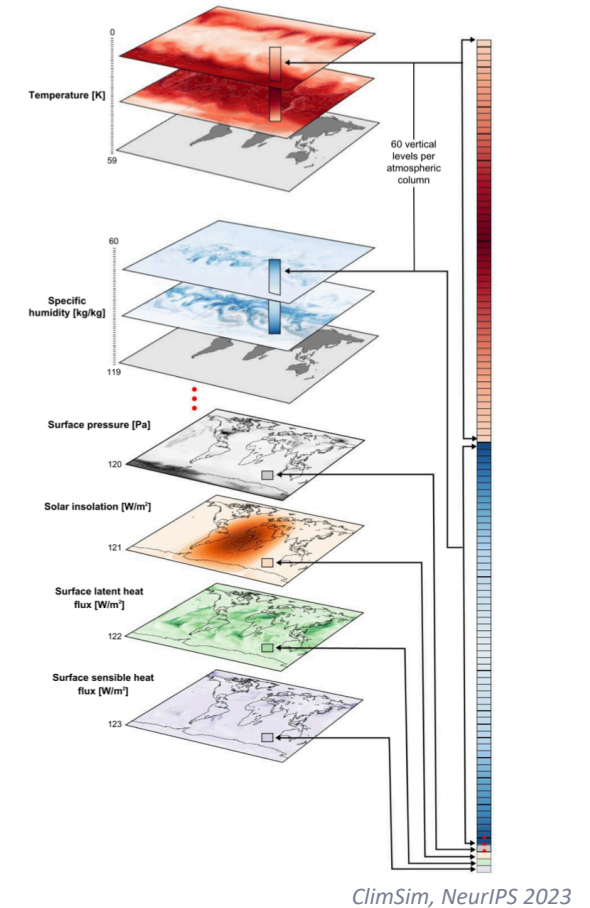
Scientific ML

Scientific ML Challenges

- **Lack of open datasets**
 - Collected from experiments or costly simulations
 - Fragmented and often proprietary
 - Small scale

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- **Physical world is complex**
 - Multi-scale and multi-physics
 - High dimensionality
 - Stochastic and partial observability



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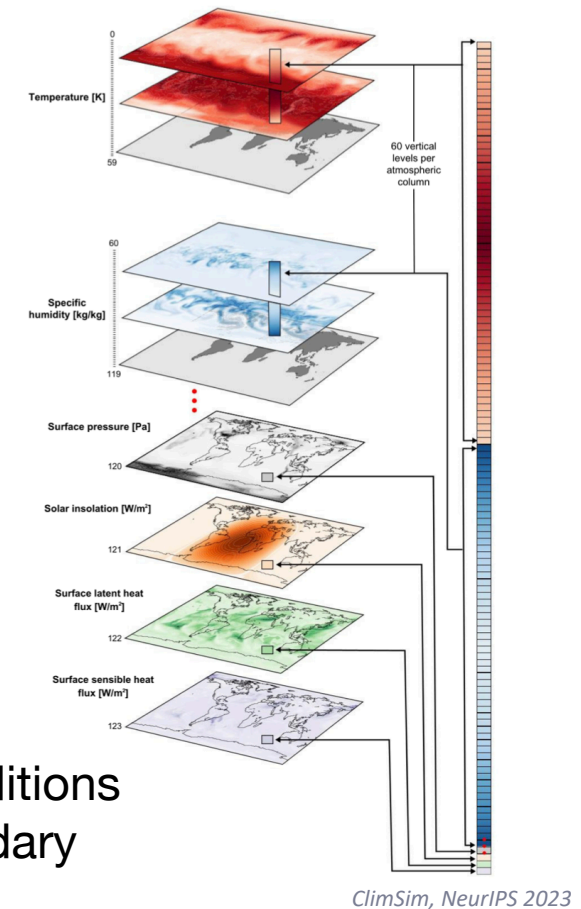
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- Physical data is scarce, models overfit to their specific training conditions
- Data-only models violate conservation laws, symmetries, and boundary conditions



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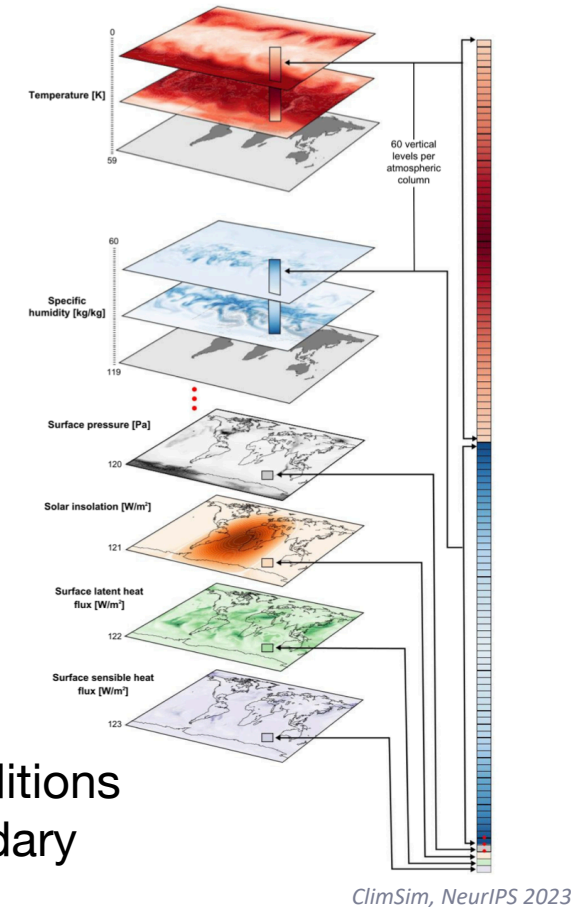
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- Data-only models violate conservation laws, symmetries, and boundary conditions

- **Interpretability**

- Must know *why* to trust the prediction to deploy safely in the real-world



Problem Formulation

A physical system is governed by a **PDE**:

$$\frac{\partial u}{\partial t} = \mathcal{N}(u, \nabla u, \nabla^2 u; \boldsymbol{\theta}), \quad x \in \Omega, t \in (0, T]$$

with **initial** and **boundary conditions**:

$$u(x, 0) = u_0(x), \quad u|_{\partial\Omega} = g$$

Define the **solution operator**:

$$\mathcal{G}_{\boldsymbol{\theta}} : \mathcal{U}_0 \longrightarrow \mathcal{U}, \quad u_0 \longmapsto u(\cdot, t)$$

$\mathbf{u}(\mathbf{x}, \mathbf{t})$ is the field we care about (velocity, temperature, pressure, ...); $\boldsymbol{\theta}$ is the physical parameter (diffusivity, viscosity, ...). Most of this course is about different ways to represent, learn, or invert $\mathbf{G}$.

Three Kinds of Problems

FORWARD

Given $\mathbf{u}_0$ and θ , predict $\mathbf{u}$. The classical simulation task, and the one surrogate models replace.

$$u = \mathcal{G}_\theta(u_0)$$

INVERSE

Given noisy observations $\mathbf{y}$, recover θ or $\mathbf{u}_0$. Often ill-posed: many causes can explain the same effect.

$$y = \mathcal{H}(\mathcal{G}_\theta(u_0)) + \varepsilon$$

DISCOVERY

Given trajectories of $\mathbf{u}$, find the governing equation $\mathcal{N}$ itself. The physics is the unknown.

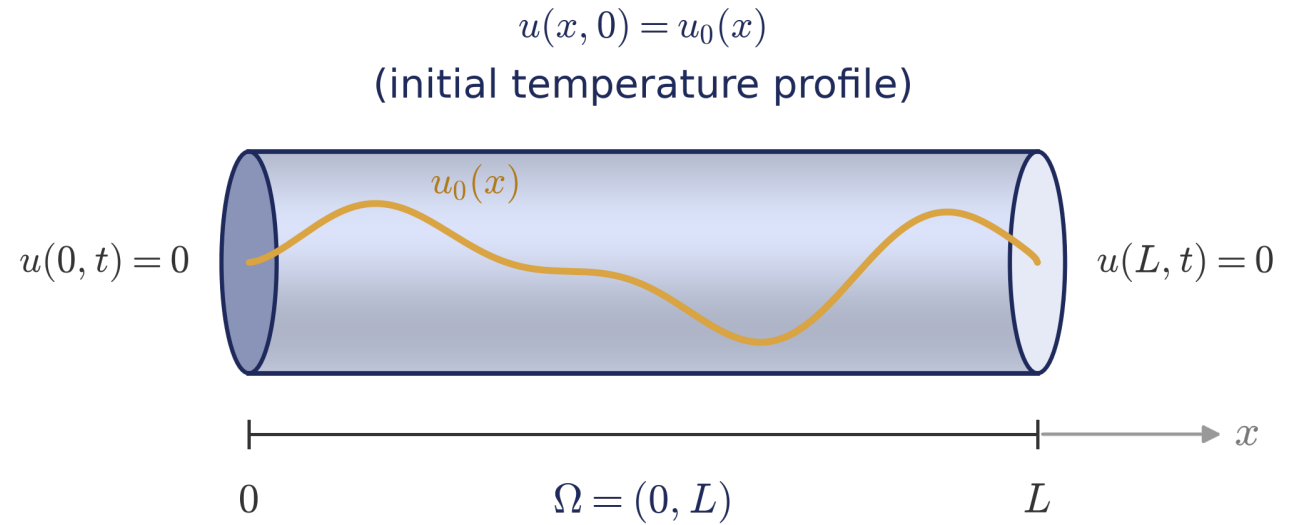
$$\frac{\partial u}{\partial t} = \hat{\mathcal{N}}(u, \nabla u, \nabla^2 u, \dots)$$

Example: Heat Equation

PDE: $\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$

IC: $u(x, 0) = u_0(x)$

BC: $u(0, t) = u(L, t) = 0$

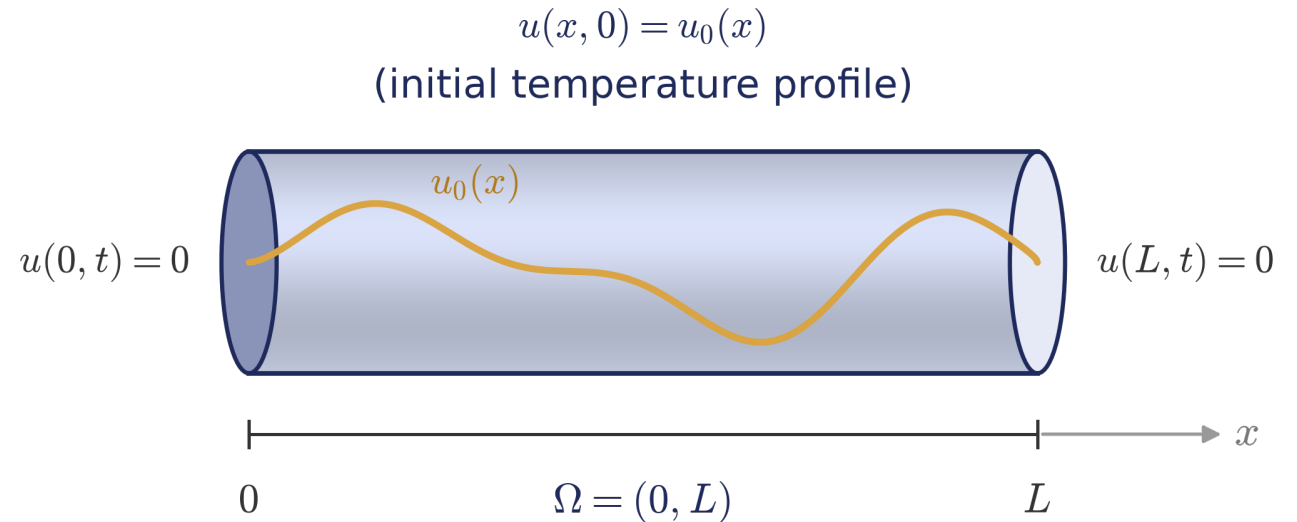


Example: Heat Equation

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FORWARD

Given $\mathbf{u}_0$ and κ , predict the temperature profile $\mathbf{u}$ at time $\mathbf{t}$.

$$u(\cdot, t) = \mathcal{G}_\kappa(u_0)$$

For this PDE, $\mathcal{G}_\kappa$ has a closed form

$$u(x, t) = \sum_{n \geq 1} c_n e^{-\kappa(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right)$$

$$c_n = \frac{2}{L} \int_0^L u_0(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

INVERSE

Given noisy sensor readings $\mathbf{y}$ at a few locations, recover κ or $\mathbf{u}_0$.

$$y_j = u(x_j, T) + \varepsilon_j$$

i.e. $y = \mathcal{H}(\mathcal{G}_\kappa(u_0)) + \varepsilon$

Recovering $\mathbf{u}_0$ means running diffusion backward, so noise in the high modes gets amplified exponentially.

mode n : $\hat{c}_n^{\text{noise}} \sim \varepsilon e^{+\kappa(n\pi/L)^2 T}$

DISCOVERY

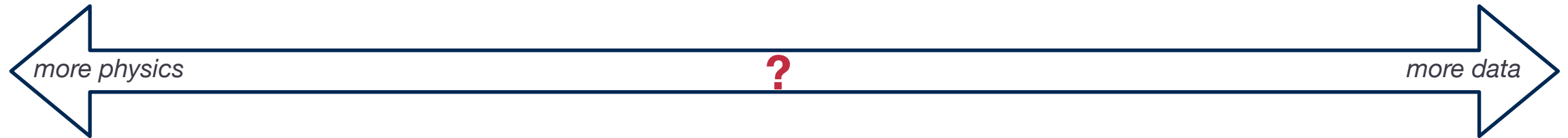
Given temperature snapshots $\mathbf{u}(\mathbf{x}_i, \mathbf{t}_k)$, find $\mathcal{N}$ from a library of candidate terms.

$$\{u(x_i, t_k)\}$$



$$\frac{\partial u}{\partial t} = \mathcal{N}(u, \nabla u, \nabla^2 u, \dots) ?$$

Physics ↔ data spectrum

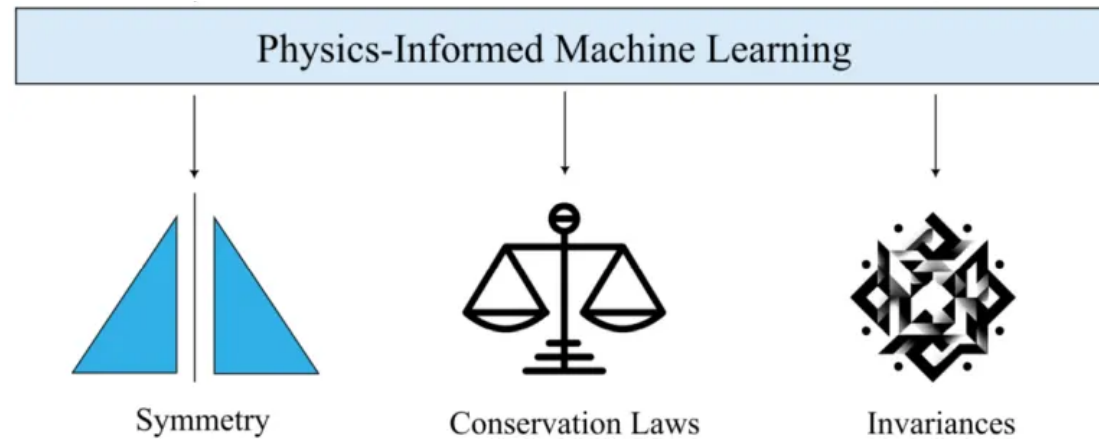


Mechanistic

Pure PDE solve
no learning

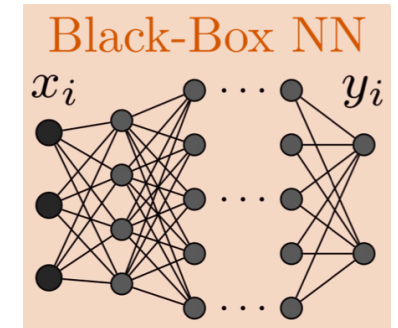
$$\frac{\partial u}{\partial t} = \mathcal{N}(u, \nabla u, \nabla^2 u; \theta)$$

$$x \in \Omega, t \in (0, T]$$



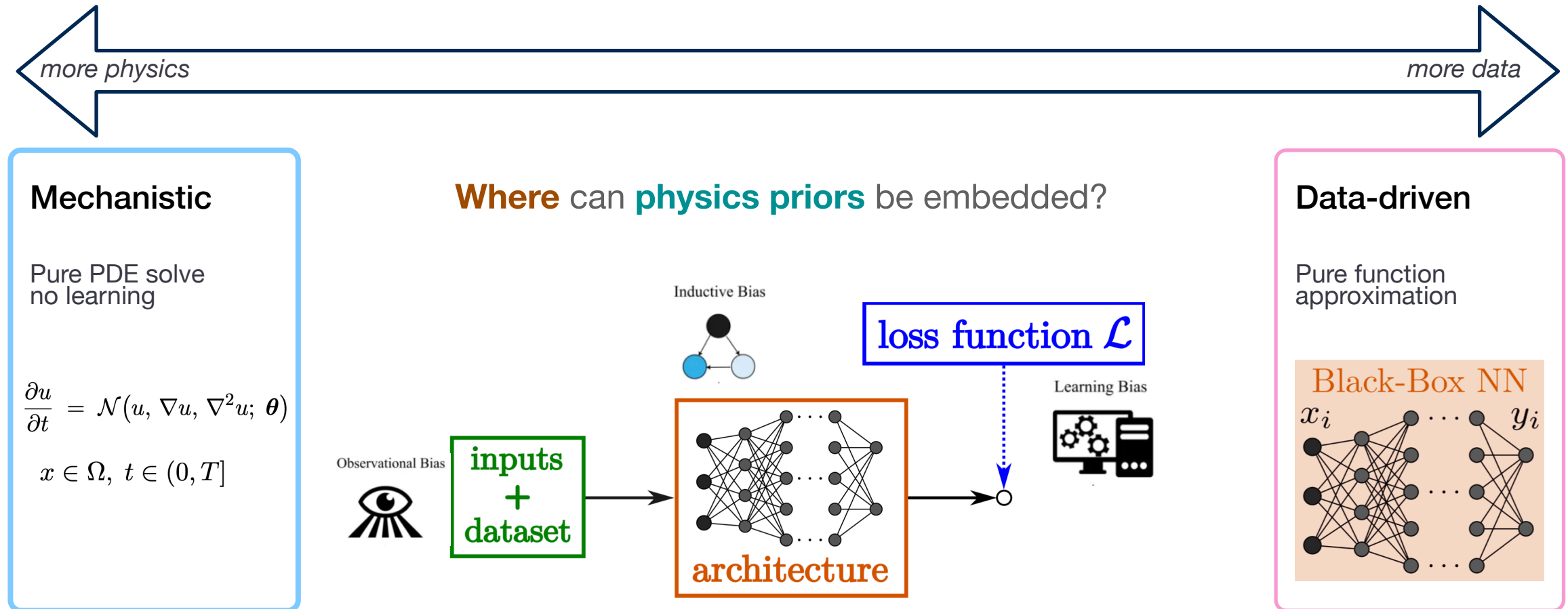
Data-driven

Pure function
approximation

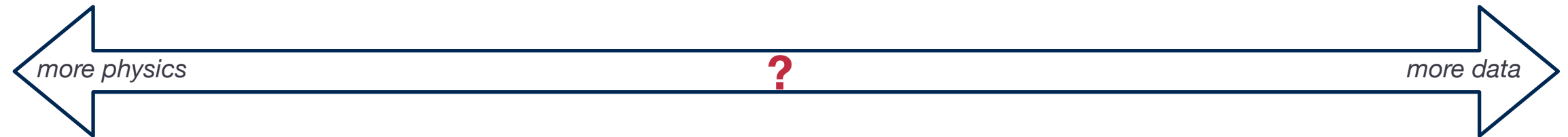


[Physics-informed machine learning](#), Nature Reviews Physics 2021

Physics ↔ data spectrum



Physics ↔ data spectrum



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$$\frac{\partial u}{\partial t} = \mathcal{N}(u, \nabla u, \nabla^2 u; \theta)$$

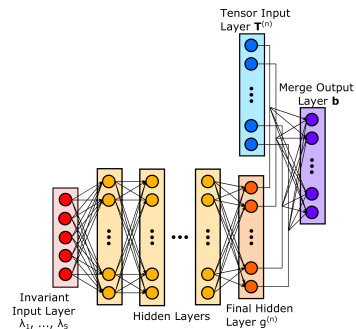
$$x \in \Omega, t \in (0, T]$$

Inductive Bias



Architecture

Neural network layers
built to satisfy
invariance by design



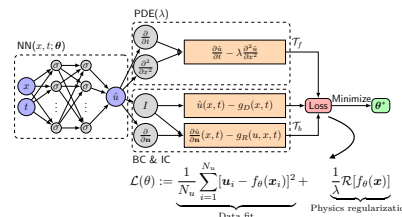
Ling et al., 2016

Learning Bias



Loss Function & Optimization

Neural network
trained to satisfy the
PDE residual



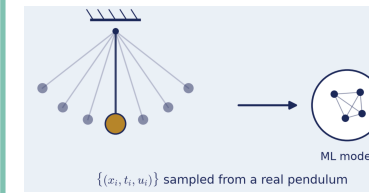
Raissi et al., 2019

Observational Bias

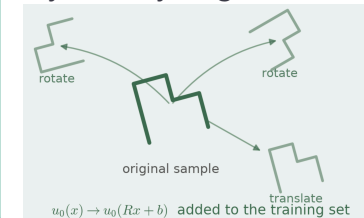


Data

Data inherits physics
from the real-world



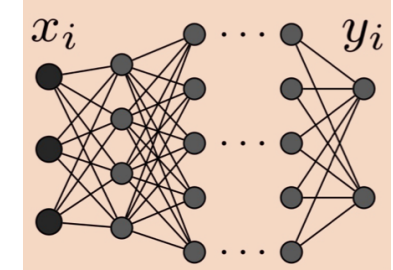
Symmetry augmentation



Data-driven

Pure function
approximation

Black-Box NN



Every method in this course sits somewhere in this spectrum. The question is never “physics or ML” but how much of each, and where.

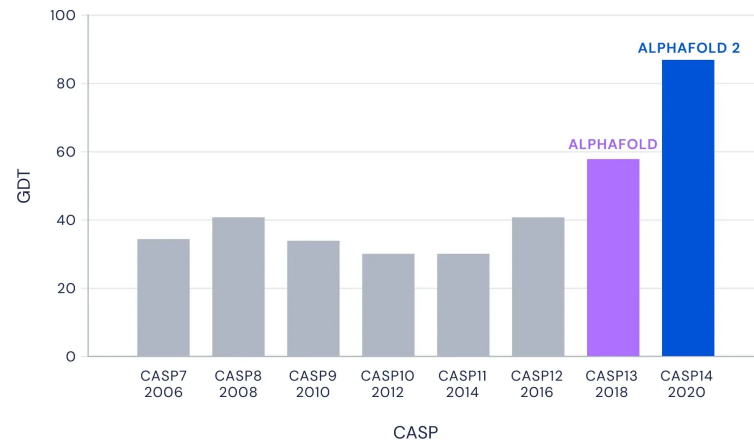
AlphaFold: Protein Structure Prediction

MIT Technology Review

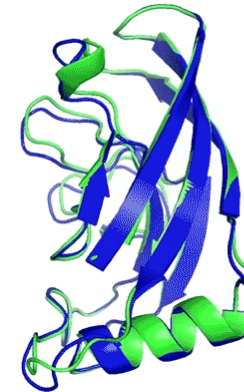
What is AlphaFold?
Proteins underpin every biological process, in every living thing.

Made from long chains of amino acids, each has a unique complex 3D structure. But figuring out just one of these can take several years, and hundreds of thousands of dollars. In 2020, AlphaFold solved this problem, with the ability to predict protein structures in minutes, to a remarkable degree of accuracy.

Median Free-Modelling Accuracy



T1037 / 6vr4
90.7 GDT
(RNA polymerase domain)



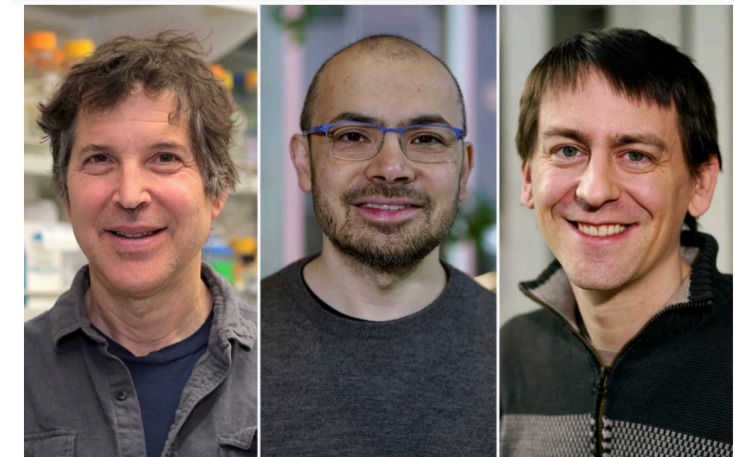
T1049 / 6y4f
93.3 GDT
(adhesin tip)

● Experimental result
● Computational prediction

DeepMind's protein-folding AI has solved a 50-year-old grand challenge of biology

AlphaFold can predict the shape of proteins to within the width of an atom. The breakthrough will help scientists design drugs and understand disease.

Chemistry Nobel goes to developers of AlphaFold AI that predicts protein structures



David Baker, Demis Hassabis and John Jumper (left to right) won the chemistry Nobel for developing computational tools that can predict and design protein structures. Credit: BBVA Foundation

For the first time – and probably not the last – a scientific breakthrough enabled by artificial intelligence (AI) has been recognized with a Nobel prize. The 2024 chemistry Nobel was

A forward map: sequence → 3D structure, learned end-to-end from data, replacing years of iterative crystallography.

GraphCast: Medium-Range Weather Forecasting

nature

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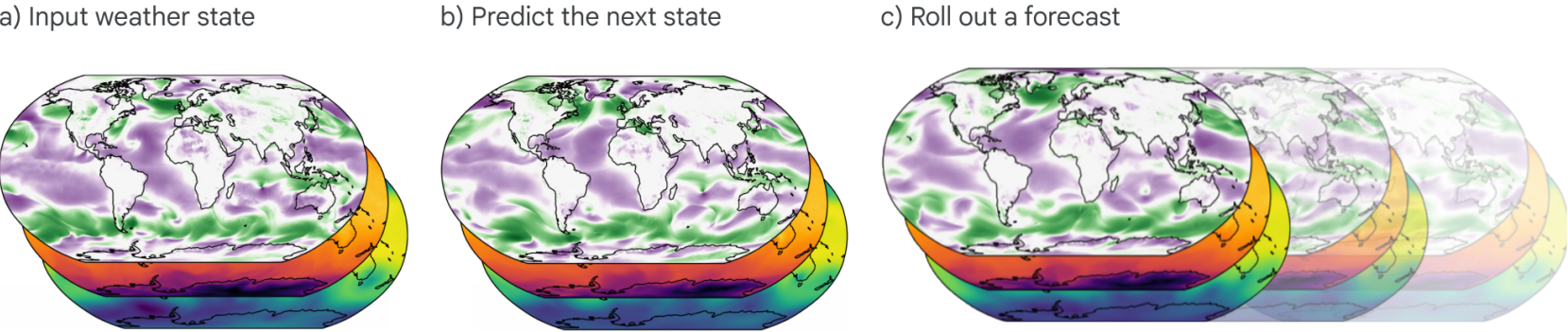
> news

> article

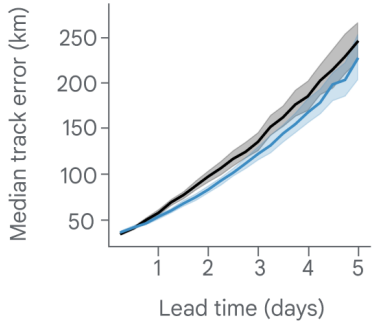
NEWS | 14 November 2023

DeepMind AI accurately forecasts weather – on a desktop computer

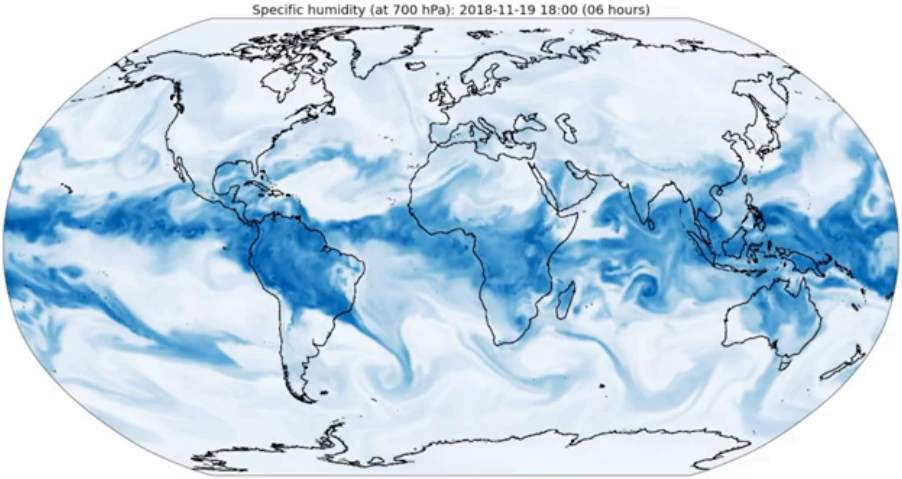
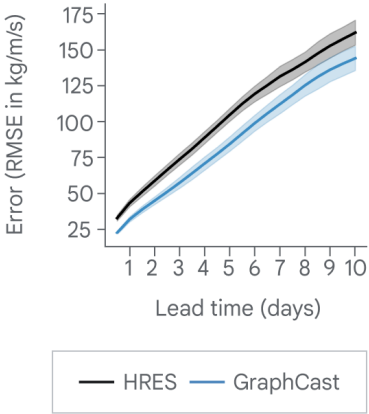
The machine-learning model takes less than a minute to predict future weather worldwide more precisely than other approaches.



Cyclone tracking

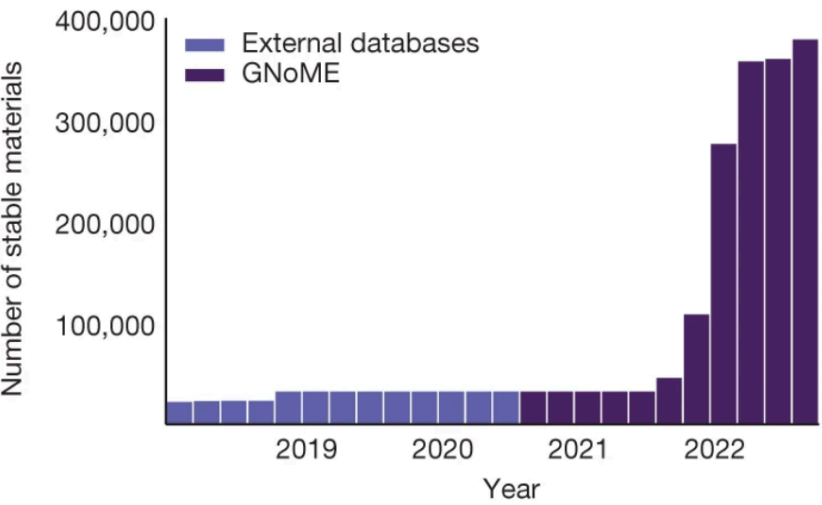


Atmospheric rivers

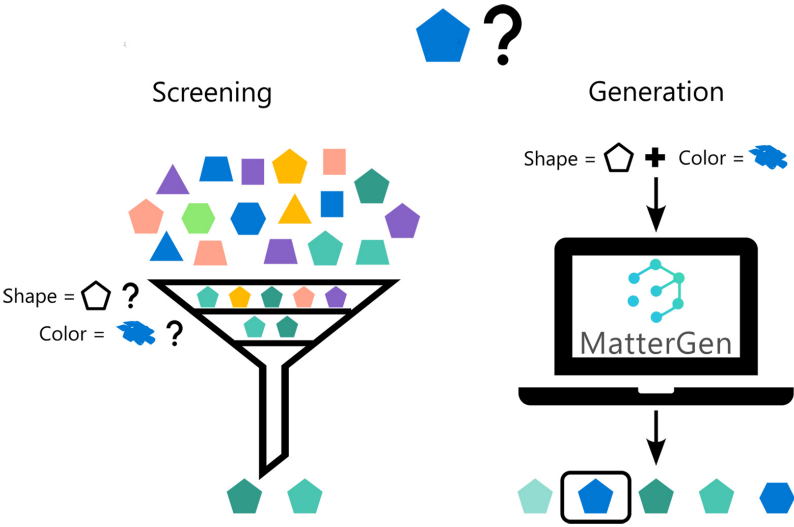


GraphCast is a graph neural network trained as a surrogate for the atmosphere's forward dynamics, at planetary scale.

AI in Materials Science



MatterGen: A new paradigm of materials design with generative AI



nature

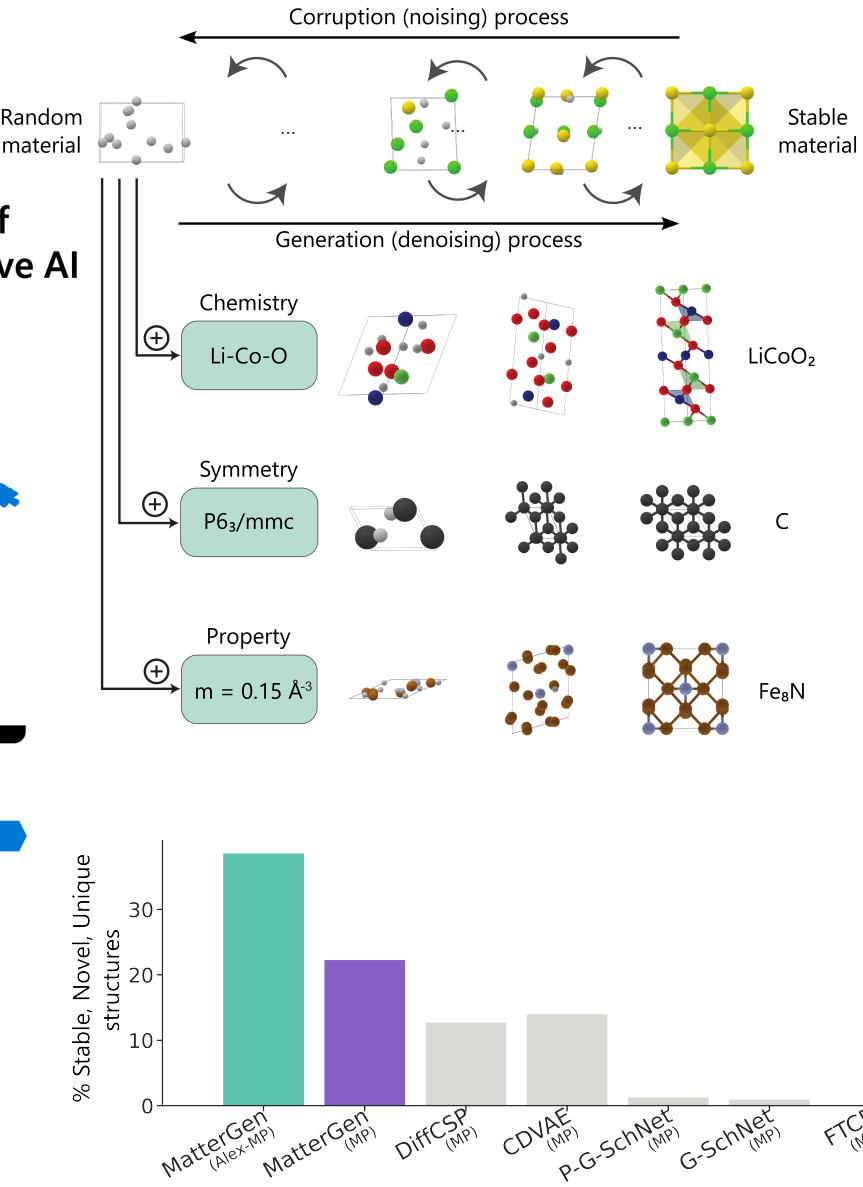
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NEWS | 29 November 2023

Google AI and robots join forces to build new materials

Tool from Google DeepMind predicts nearly 400,000 stable substances, and an autonomous system learns to make them in the lab.

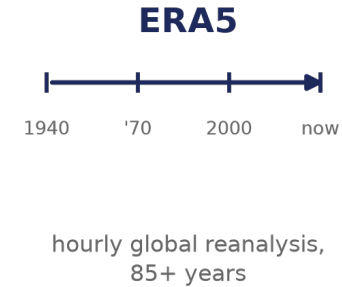


Merchant et al., "Scaling deep learning for materials discovery," Nature 2023.
Zeni et al., "A generative model for inorganic materials design," Nature 2025.

Why is this possible now?

- **Data at scale**

- Decades of reanalysis, simulation, and experimental data (ERA5, Materials Project, PDB) generated large training data.



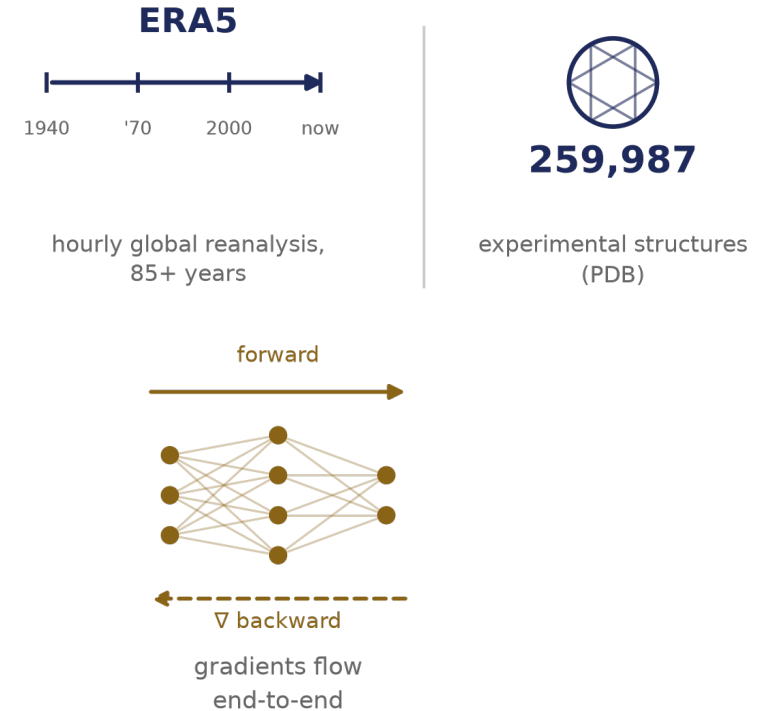
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- Automatic differentiation and GPU compute made it possible to train models with hundreds of millions of parameters, end-to-end.



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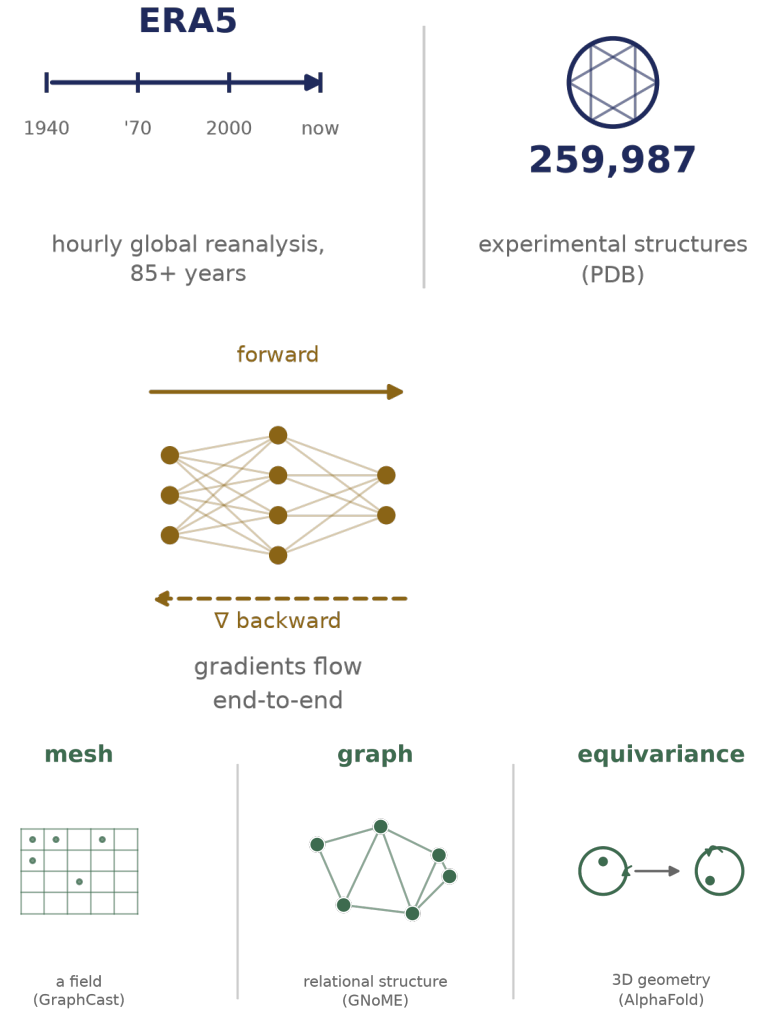
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● Architectures matched to inputs

- Each architecture encodes the shape of its data: mesh for a field, graph for structure, equivariance for geometry.



Open Problems

**Rollout stability
over long horizons**



**Generalizing across
resolution & geometry**



**Hard constraints
vs. expressivity**



**Calibrated
uncertainty**



**Do foundation models
generalize across domains?**



Part I
What's
different?

Part II
Spatial &
temporal

Part III
Operators &
graphs

Part IV
Physics-
informed

Part V
Generative
& UQ

Part VI
Scaling &
foundation

Course Logistics

ML4SCI: Machine Learning for Science

<https://eecs-ml4sci.github.io>

University of California, Irvine · Fall 2026

[Course details](#) [Schedule](#) [Course FAQs](#)

Meets	Tue / Thu, 8:00–9:20 AM · ICS 213
Format	~60 min lecture + ~20 min live notebook demo
Prerequisites	Deep learning fundamentals (GD, backprop, training, etc) assumed PDE and numerical methods basics (FD, FV, FEM, iterative solvers, etc) assumed
Grading & policies	25% Participation (Expected to attend all lectures and engage with the course materials. For notebook-guided lectures, complete the exercises and submit the notebook on Canvas) 75% Final project
Project	Conduct an original investigation, replicate and critically evaluate a recent paper, extend an existing method, or compare modeling approaches in machine learning for science

Course Content

Part I — What makes ML for Science different?		
1	Thu, Sep 24	Introduction
2	Tue, Sep 29	ML Foundations for Scientific Data
3	Thu, Oct 1	PDE and Numerical Fundamentals
4	Tue, Oct 6	Data Representations for SciML
5	Thu, Oct 8	Symmetry, Invariance, and Equivariance
Part II — Learning spatial and temporal dynamics		
6	Tue, Oct 13	CNN and U-Net Architectures for Spatial Fields
7	Thu, Oct 15	Autoregressive Rollout Models
8	Tue, Oct 20	Temporal Modeling: Recurrent Networks and Neural ODEs
9	Thu, Oct 22	Temporal Modeling: Attention and Transformers
10	Tue, Oct 27	Evaluating SciML

Part III — Changing the computational formulation		
11	Thu, Oct 29	Operator Learning
12	Tue, Nov 3	GNNs and Mesh-Based Learning
Part IV — Incorporating scientific knowledge		
13	Thu, Nov 5	Physics-Informed ML
14	Tue, Nov 10	Physics-Constrained and Hybrid Learning
Part V — From predictions to distributions		
15	Thu, Nov 12	Generative and Probabilistic Models for Science
16	Tue, Nov 17	Uncertainty Quantification
Part VI — Scaling scientific intelligence		
17	Thu, Nov 19	Scaling Laws for SciML
18	Tue, Nov 24	Mixture-of-Experts and Scientific Foundation Models
19	Thu, Nov 26	No Class — Thanksgiving
19	Tue, Dec 1	Mechanistic Interpretability for Science
20	Thu, Dec 3	Course Synthesis

From concepts to code

In this notebook, we'll load two small **PDE datasets**, look at what makes it different from an image or a token sequence, and set up the **forward problem** baseline

→ [Open the Lecture 1 notebook](#)